Accurate and Simple: Length-Penalized Detection Scoring for Sparse Autoencoder Feature Explanations

Libor "Burny" Burian

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Abstract

Detection scoring is the standard way to grade natural-language explanations of sparse autoencoder (SAE) latents (Paulo et al. 2024; SAEBench). It rewards accuracy only. We propose the smallest change that also rewards brevity, $S = \max(J, 0) \cdot 2^{-L/H}$: Youden's $J = 2\,\mathrm{BA} - 1$ from the detection scorer, times a factor that halves every H tokens of explanation length L.

We test it with no training and minimal custom code:

- **Data:** Neuronpedia's public explanations and activations for 200 randomly sampled latents of a GPT-2 small residual SAE, plus random negatives verified with the same SAE.
- **Code:** delphi's unmodified detection scorer and explainer, with Llama-3.1-70B-Instruct.

We pre-registered four hypotheses. Results:

- The scorer is valid. Random explanations score J = -0.022; real ones 0.444–0.499.
- Asking the same explainer for ~2 to ~40 words raises J from 0.464 to 0.577, with diminishing returns.
- Our primary hypothesis, an interior optimum of S at the pre-registered $H = 20$, is **falsified**. Mean S decreases monotonically with length, so on average the shortest explanations win. In an exploratory sweep, ~10-word explanations win only at a much weaker penalty ($H = 80$).
- Used to choose among 8 candidate explanations per latent, with cross-fitted held-out evaluation, S selects explanations 39% shorter than plain detection scoring does, at a cost of 0.031 J [0.018, 0.046]. That is 0.146 J better than simply picking the shortest.

A one-token baseline (`the token "X"`) is as good under S as gpt-4o-mini's explanations. Code, data, logs and pre-registration: https://github.com/BurnyCoder/accurate-and-simple-mechanistic-interpretability-explanations.

1. Introduction

Sparse autoencoders (SAEs) decompose a language model's activations into many sparse latents ("features"), and each latent is usually given a natural-language explanation written by an LLM. Explanations are judged by *auto-interpretability* scores. An LLM is given only the explanation and asked to predict where the latent fires. Two ways of doing this are simulating per-token activations (Bills et al. 2023) and classifying whole contexts as activating or not, called detection (Paulo et al. 2024). Detection scoring is the basis of SAEBench's autointerp metric (Karvonen et al. 2025) and is implemented in EleutherAI's open-source delphi library. A recent study of autointerp pipelines found it to be the most stable autointerp metric (van der Ben et al. 2026).

These scores reward accuracy only. Paulo et al. state that "current scoring methods do not account for interpretation length". An explanation that lists every pattern it noticed is not penalized for its length, even though a human reader, or a downstream circuit analysis, pays for every word. LLM judges in other settings are known to prefer longer outputs (Saito et al. 2023), and leaderboards have had to correct for length (Dubois et al. 2024).

We make the smallest change that puts brevity into the objective. We multiply the above-chance detection score by a factor that halves every H tokens:

$$S = \max(J, 0) \cdot 2^{-L/H}, \qquad J = 2\,\mathrm{BA} - 1.$$

Here BA is delphi's balanced accuracy and L is the explanation's length in tokens. In log space, $\log S = \log J - (\ln 2/H)\, L$: a constant cost per token. This is the two-part code of minimum description length (Rissanen 1978), applied to the text explanation rather than to the SAE code as in Ayonrinde et al. 2024.

To keep custom work minimal, we use existing explanations and activation data from Neuronpedia's public export for GPT-2 small, and delphi's scorer and explainer unchanged.

2. Background: detection scoring

For a latent and an explanation, the scorer LLM sees contexts in shuffled batches of 5. Some contexts contain the latent's activation and some do not. For each context it answers 1 or 0 ("does this context match the explanation?"). Paulo et al. use 100 activating contexts (10 per activation decile) and 100 random non-activating contexts, with Llama-3.1-70B-Instruct (4-bit) as both explainer and scorer. Because non-activating contexts are included, the score reflects both recall (do activating contexts match?) and precision (do non-activating ones not match?) of the explanation.

We use delphi's `DetectionScorer` with its default prompt and three few-shot examples. Two of those examples are latents of the very SAE we study (6048 and 9396), so we exclude them.

3. Method

Score. delphi reports balanced accuracy BA = (TPR + TNR)/2, computed with its own `compute_confusion` and `compute_classification_metrics`. We use Youden's $J = 2\,\mathrm{BA} - 1 = \mathrm{TPR} - \mathrm{FPR}$, because chance is then 0. If the length factor multiplied BA instead, a short random explanation (BA ≈ 0.5) would beat a long accurate one. The clip $\max(J, 0)$ keeps below-chance explanations from gaining by being short.

L is the token count of the stripped explanation under tiktoken's `o200k_base` encoding. That tokenizer is fixed, so L does not depend on the scorer. Words and characters are checked as alternatives.

The half-life H is the exchange rate between accuracy and brevity. We pre-registered H = 20 tokens: a typical 10-token explanation keeps 71% of its J and a 40-token one keeps 25%. We report a sweep over H.

Data.

- **SAE and latents.** We use Joseph Bloom's residual-stream SAE for GPT-2 small at layer 6 (`blocks.6.hook_resid_pre`, 24,576 ReLU latents), as exported by Neuronpedia. A latent is eligible if it has explanations from both gpt-3.5-turbo and gpt-4o-mini, at least 45 activating 17-token contexts, and is not one of delphi's few-shot latents. That leaves 17226 of 24576 (70.1%). We sample 200 at random (seed 0).
- **Contexts per latent.**
  - The 20 strongest contexts are reserved for the explainer.
  - 25 test positives are drawn, 5 from each of 5 equal-count activation-quantile bins of the remaining contexts.
  - 25 negatives are drawn from 4000 random OpenWebText windows on which the latent's activation is exactly 0.
- **Verifying the negatives.** We compute those activations with sae_lens and TransformerLens on GPT-2 small. Re-running Neuronpedia's top windows reproduces its exported maxima (Pearson r = 0.988, n = 152). Negatives are displayed in Neuronpedia's exact token format (100% round-trip), so the format cannot reveal the label.
- **Why not Neuronpedia's own zero rows.** Neuronpedia's zero-activation rows were our pre-registered negatives. A pilot showed they are near-duplicate sliding windows: 5250 rows but 2010 distinct texts. Their replacement was recorded as a deviation before any main results (Appendix A).

Conditions.

- **Experiment 1:**
  - Neuronpedia's gpt-3.5-turbo and gpt-4o-mini explanations;

  - a top-token baseline (`the token "X"`, X the most frequent peak token among the train contexts);
  - a random control (another sampled latent's explanation).
- **Experiment 2:** Llama-3.1-70B-Instruct writes six explanations per latent with delphi's default explainer prompt, shown the 20 train contexts. One sentence is appended asking for approximately $B \in \{2, 5, 10, 20, 40\}$ words, plus one unconstrained explanation.

Scoring. delphi `DetectionScorer`, Llama-3.1-70B-Instruct via OpenRouter (pinned to DeepInfra fp8), temperature 0, 5 contexts per prompt. All conditions of a latent are scored on the same 50 contexts in the same shuffled batches. 2002 explanations were scored with 21220 calls (US$6.78). 0.035% of samples were lost to unparseable answers.

Statistics. Means over latents with 95% percentile bootstrap CIs (10,000 resamples), paired one-sided Wilcoxon signed-rank tests, and Holm correction within each hypothesis.

For H4, each latent's 50 test contexts are split into class-stratified halves. A candidate is selected on one half and its J measured on the other, then the halves are swapped. Selection and evaluation therefore never share contexts (cross-fitting).

4. Results

4.1 The detection scorer is valid (H1)

explanation source	role	n	mean BA	mean J [95% CI]	median tokens	mean S (H=20) [95% CI]
random-explanation control	control	200	0.489	-0.022 [-0.038, -0.006]	8	0.021 [0.015, 0.029]
top-token baseline	baseline	200	0.715	0.430 [0.383, 0.476]	5	0.365 [0.326, 0.403]
gpt-3.5-turbo (Neuronpedia)	existing	200	0.722	0.444 [0.399, 0.490]	8	0.341 [0.307, 0.377]
gpt-4o-mini (Neuronpedia)	existing	200	0.750	0.499 [0.452, 0.546]	8	0.374 [0.339, 0.408]
Llama-3.1-70B, ∼2 words	Exp 2	200	0.732	0.464 [0.422, 0.506]	2	0.430 [0.393, 0.467]
Llama-3.1-70B, ∼5 words	Exp 2	200	0.747	0.493 [0.455, 0.530]	6	0.395 [0.365, 0.425]
Llama-3.1-70B, ∼10 words	Exp 2	200	0.774	0.548 [0.509, 0.588]	11	0.374 [0.348, 0.400]
Llama-3.1-70B, ∼20 words	Exp 2	200	0.776	0.552 [0.513, 0.590]	20	0.278 [0.258, 0.297]
Llama-3.1-70B, ∼40 words	Exp 2	200	0.789	0.577 [0.541, 0.613]	45	0.123 [0.115, 0.131]
Llama-3.1-70B, delphi default prompt	Exp 2	200	0.781	0.563 [0.524, 0.601]	30	0.204 [0.189, 0.220]

Table 1: All conditions. BA = balanced accuracy. 95% CIs in brackets.

The random control scores J = -0.022 [-0.038, -0.006]. That is within the pre-registered $|J| < 0.05$, but its CI lies slightly below zero. Neuronpedia's explanations score well above chance. gpt-4o-mini's are more accurate than gpt-3.5-turbo's by 0.055 [0.014, 0.096] (two-sided p = 0.065, exploratory). Both have a median of 8 tokens, so S keeps their order.

The one-token baseline reaches J = 0.430. Under S it scores 0.365, level with gpt-4o-mini (0.374) and above gpt-3.5-turbo (0.341).

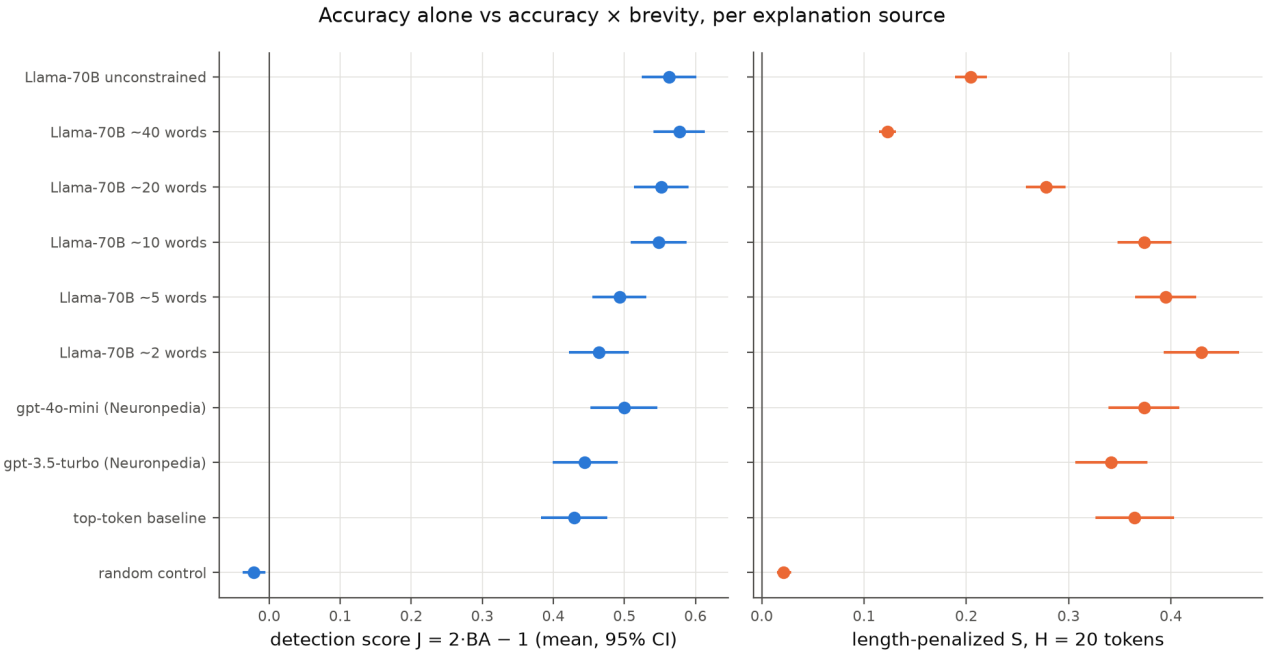


Figure 1: Mean J (left) and S at $H = 20$ (right) per explanation source, with 95% CIs.

4.2 Accuracy rises with length, with diminishing returns (H2)

Asked for ~2, ~5, ~10, ~20 and ~40 words, the explainer produced median lengths of 2, 6, 11, 20 and 45 tokens. Mean J was 0.464, 0.493, 0.548, 0.552 and 0.577.

- **Rise:** $J(40) - J(2)$ = 0.113 [0.079, 0.147], p = 7.1e-09.
- **Diminishing returns:** the gain per word from 2→10 words exceeds that from 10→40 by 0.0095 [0.0056, 0.0136].

H2 is supported.

4.3 No interior optimum at $H = 20$ (H3, primary): falsified

comparison (S at H=20)	mean difference [95% CI]	Wilcoxon p (one-sided)	Holm p
~5 words – ~2 words	-0.035 [-0.064, -0.006]	0.99	1
~5 words – ~40 words	+0.272 [+0.246, +0.297]	6.6e-33	3.3e-32
~10 words – ~2 words	-0.056 [-0.081, -0.031]	1	1

comparison (S at H=20)	mean difference [95% CI]	Wilcoxon p (one-sided)	Holm p
∼10 words − ∼40 words	+0.251 [+0.230, +0.271]	1.9e-33	1.1e-32
∼20 words − ∼2 words	-0.152 [-0.181, -0.123]	1	1
∼20 words − ∼40 words	+0.155 [+0.141, +0.169]	6.8e-33	3.3e-32

Table 2: Paired differences in S ($H = 20$) between budgets.

Mean S is 0.430, 0.395, 0.374, 0.278 and 0.123 for ∼2 to ∼40 words. Every intermediate budget beats ∼40 words, but none beats ∼2 words, so H3 is falsified.

The reason is quantitative. From ∼2 to ∼10 words mean J grows by a factor of 1.18, while the length factor shrinks by 0.73.

In an exploratory, post-hoc sweep (Table 3, Fig. 4) the best budget is ∼2 words for $H \leq 20$. At $H = 40$, ∼10 words and ∼2 words tie: +0.005 [-0.022, +0.032]. At $H = 80$, ∼10 words beats ∼2 words by 0.041 [0.013, 0.069] and ∼40 words by 0.108 [0.087, 0.128]. These comparisons were chosen after seeing the data.

H (tokens)	random-explanation control	top-token baseline	gpt-3.5-turbo (Neuronpedia)	gpt-4o-mini (Neuronpedia)	Llama-3.1-70B, ∼2 words	Llama-3.1-70B, ∼5 words	Llama-3.1-70B, ∼10 words	Llama-3.1-70B, ∼20 words	Llama-3.1-70B, ∼40 words	Llama-3.1-70B, delphi default prompt	best budget
5	0.009	0.216	0.161	0.163	0.332	0.203	0.119	0.037	0.001	0.014	∼2 words
10	0.016	0.306	0.262	0.280	0.394	0.316	0.255	0.140	0.027	0.079	∼2 words
20	0.021	0.365	0.341	0.374	0.430	0.395	0.374	0.278	0.123	0.204	∼2 words
40	0.025	0.398	0.391	0.434	0.449	0.442	0.454	0.392	0.266	0.337	∼10 words
80	0.027	0.416	0.419	0.468	0.459	0.467	0.500	0.467	0.392	0.435	∼10 words
∞ (no length factor: max(J, 0))	0.029	0.435	0.450	0.505	0.469	0.494	0.551	0.556	0.578	0.565	∼40 words

Table 3: Mean S per condition for each half-life H (exploratory).

4.4 Selecting by S gives short and accurate explanations (H4)

selection rule	held-out mean J [95% CI]	median tokens
pick highest J (plain detection)	0.644 [0.605, 0.682]	9.0
pick highest S (this method)	0.613 [0.574, 0.649]	5.5
pick shortest	0.467 [0.425, 0.509]	2.0

selection rule	held-out mean J [95% CI]	median tokens

Table 4: Cross-fitted selection among 8 candidates per latent.

- Selecting by S instead of J shortens the chosen explanation by 39% (median 5.5 vs 9.0 tokens).
- It loses 0.031 held-out J.
- It beats choosing the shortest candidate by 0.146 [0.115, 0.179].

All three pre-registered criteria hold. Selection by S almost never picks the ∼40-word or unconstrained explanations: 2 and 1 picks out of 400, against 28 and 29 under J.

4.5 Reliability and robustness

re-scoring	latents	explanations	Spearman J	Spearman S
same scorer, new shuffle seed (43)	50	502	0.961	0.957
Llama-3.3-70B as scorer	50	502	0.971	0.968

Table 5: Re-scoring 50 latents (all conditions).

Measuring L in words or characters instead of tokens leaves the ranking of explanations by S almost unchanged: Spearman 0.998 and 0.996, with H rescaled by the corpus words-per-token and characters-per-token ratios.

5. Discussion

What the length factor does well. As a *selection* criterion over candidate explanations of the same latent, S achieves the intended trade-off: much shorter explanations at a small, measured loss in accuracy. It is also robust:

- Short-but-wrong explanations cannot win: the random control has $S \approx 0$.
- Selecting by S is far better than simply preferring brevity.

What it does not do at $H = 20$. Averaged over latents, the pre-registered penalty is harsh enough that the shortest explanations win. For GPT-2 layer-6 latents, a 2-word label already captures most of the signal detection scoring can measure: J 0.464 vs 0.577 at ∼40 words. Two readings fit this result:

- many of these latents are close to single-token detectors, in line with the strong top-token baseline;
- detection on 17-token contexts cannot reward nuance beyond what a short label carries.

Choosing H. H is the exchange rate between accuracy and brevity, the price in tokens of halving the score. It is a value judgement and should be set for the use case. In our sweep, labels win for $H \leq 40$, and ∼10-word explanations need $H \approx 80$. Because $\log_2 S = \log_2 J - L/H$, H also has an MDL reading: each token costs $1/H$ bits of $\log_2 J$.

6. Limitations

- **One SAE and one scorer family.** GPT-2 small, layer 6, with Llama 70B scorers. The reliability checks cover a second scorer on 50 latents only.
- **Short contexts and a small test set.** Contexts are 17 tokens, and each explanation is scored on 50 contexts rather than Paulo et al.'s 200, so single-explanation scores are noisy (Table 5). Our claims are about means over 200 latents.
- **Possible overlap.** Neuronpedia's explanations were written from top activations. Holding out the 20 strongest contexts reduces but does not remove overlap with our test positives. The Exp 2 explanations are fully held out.
- **Sample bias.** The eligibility filter (≥ 45 activating contexts) biases the sample toward denser latents: 70.1% of latents are eligible.
- **Tokenizer dependence.** L depends on the tokenizer, but rankings barely change with words or characters.

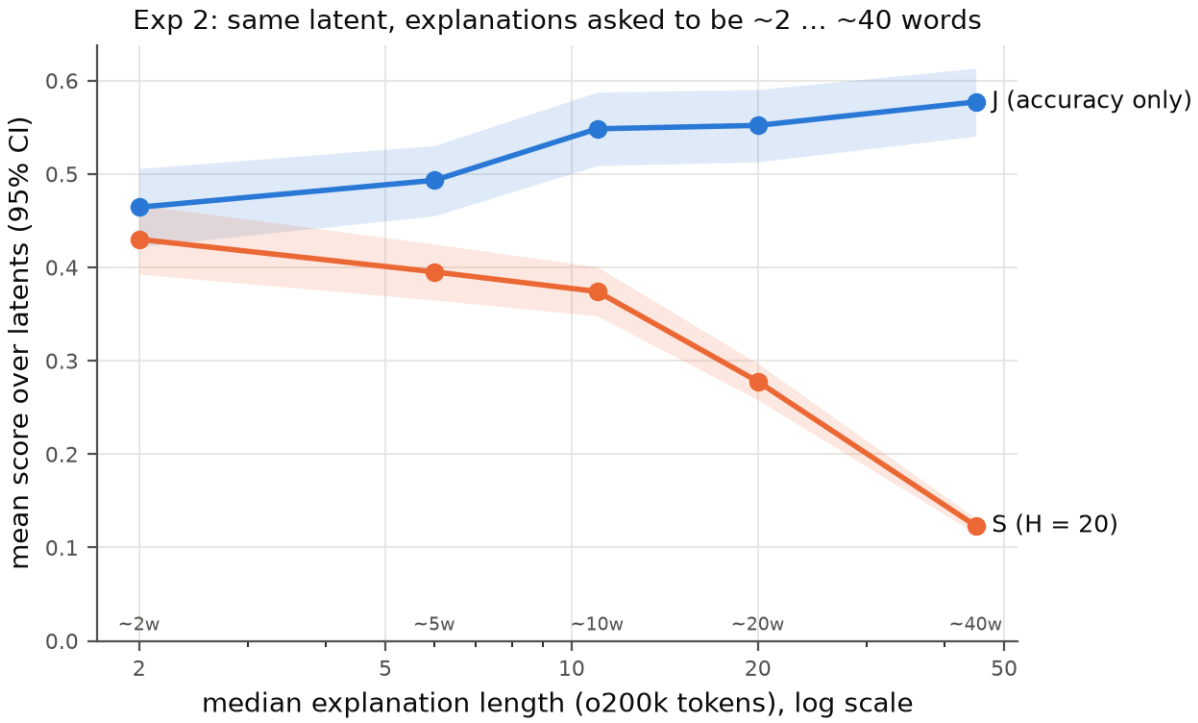


Figure 2: Mean J and S ($H = 20$) against median explanation length in Experiment 2.

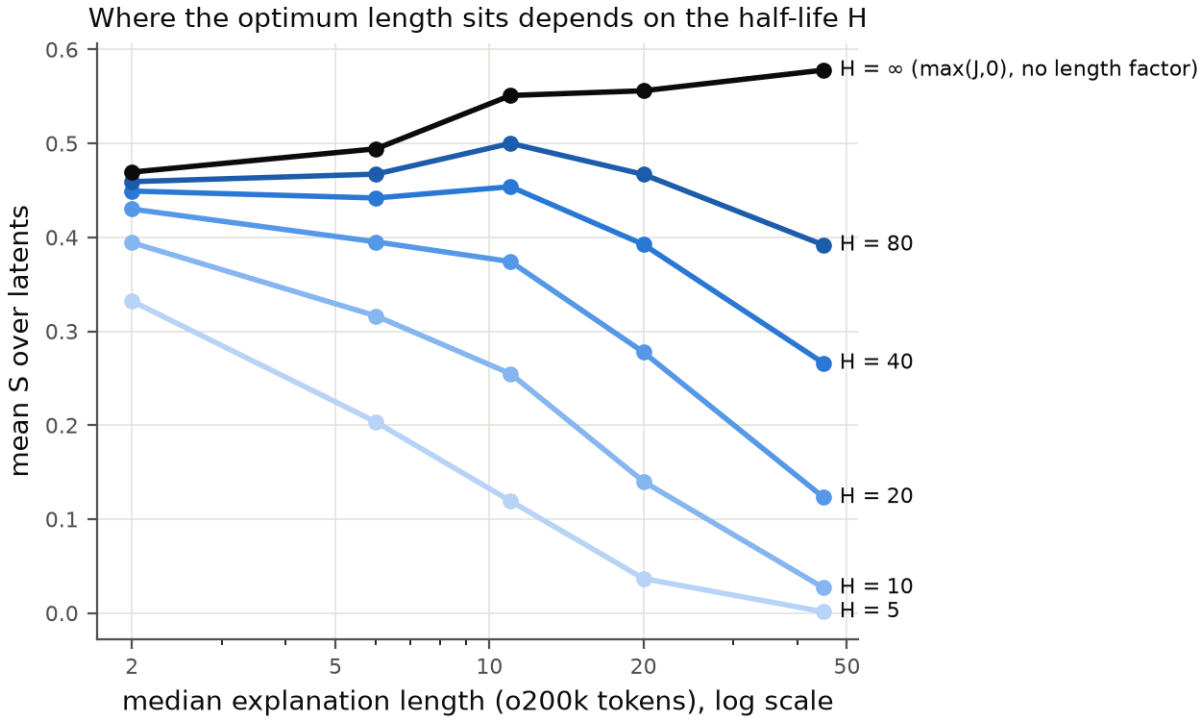


Figure 3: Mean S per length budget for each H.

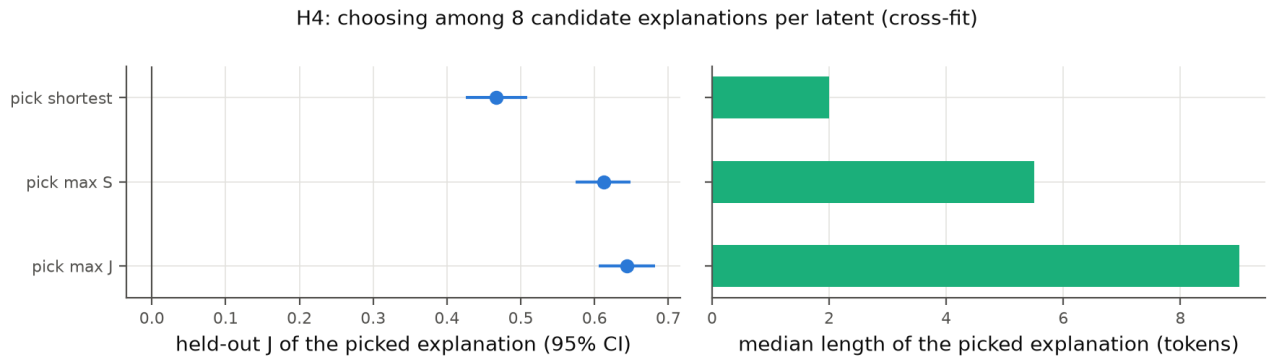


Figure 4: Held-out J and median length of the explanation chosen by each rule.

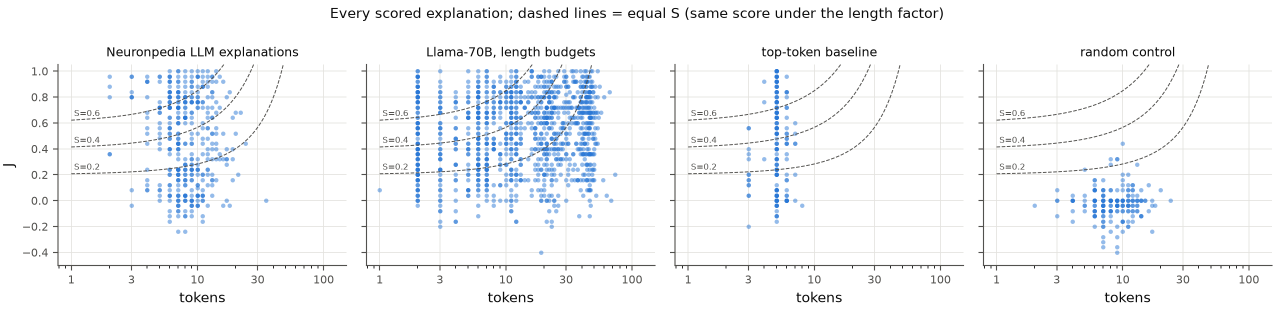


Figure 5: Every scored explanation: J against length; dashed curves have equal S.

- **Untested properties.** We did not test whether shorter explanations are more *useful* to humans or downstream tools. We assume it, as the premise of the project.

7. Conclusion

Multiplying the above-chance detection score by $2^{-L/H}$ is a one-line, library-compatible way to make SAE explanation scores reward brevity. Used to choose explanations, it gives much shorter ones at a small cost in accuracy. The pre-registered strength $H = 20$ is, however, strong enough that on average the shortest explanations win, so H should be chosen deliberately; in our post-hoc sweep, ~10-word explanations win only at $H \approx 80$.

Related work

Auto-interpretability scoring.

- Bills et al. (2023) introduced explain-then-simulate scoring for neurons.
- Paulo et al. (2024) introduced cheaper scorers (detection, fuzzing, surprisal, embedding, intervention) and the delphi library. They note that "current scoring methods do not account for interpretation length".
- SAEBench (Karvonen et al. 2025) adopted their LLM-as-judge detection metric.
- van der Ben et al. (2026) show that pipeline choices dominate autointerp score variance, with detection the most stable metric.

None of these scores penalizes length. Our score is a one-line extension of detection that does.

Simplicity and description length.

- The minimum description length principle (Rissanen 1978) scores a hypothesis by the bits needed to state it plus the bits needed to encode the data given it.

- Ayonrinde, Pearce and Sharkey (2024) apply MDL to the SAE's latent code, seeking explanations that are "both accurate and concise".
- Ayonrinde and Jaburi (2025) list explanatory simplicity among the virtues of mechanistic explanations.

We apply the same fit-plus-length trade-off to the natural-language explanation itself, with a single interpretable exchange rate H.

Length bias in LLM evaluation.

- LLM judges prefer longer answers (Saito et al. 2023).
- Length-controlled AlpacaEval removes this bias by regression (Dubois et al. 2024).

Our goal is the opposite of length *control*. We deliberately *reward* brevity, because shorter explanations are cheaper for humans and downstream tools to use.

Data.

- The SAEs are Bloom (2024)'s open residual-stream SAEs for GPT-2 small.
- Explanations and dashboards are from Neuronpedia.
- Random contexts are from OpenWebText.
- The SAE is loaded with sae_lens.

Appendix A: Pre-registration and deviations

Hypotheses, metrics, tests and falsification criteria were committed in `PREREG.md` before any results. Two deviations were made before any main-sample results.

1. The Exp 2 budget sentence asks for "approximately B words" instead of "at most B words", because delphi's explainer prompt already demands concision.
2. The negatives changed from Neuronpedia's zero rows to verified random OpenWebText windows, after the 10-latent debugging pilot showed the zero rows were near-duplicates.

The analysis plan was not changed. The H sweep tests are labelled exploratory.

Appendix B: Prompts

- **Scorer:** delphi's detection prompt (`delphi/scorers/classifier/prompts/detection_prompt.py`, commit `4fea06e`). It has a system prompt, three few-shot examples from this SAE (two of them latents 6048 and 9396, which we exclude), and the test batch.
- **Explainer:** delphi's `DefaultExplainer` prompt (`delphi/explainers/default/prompts.py`), with "Your final explanation (the [EXPLANATION]: line) must be approximately B words long." appended to the last user message.

Every prompt and response is in the run logs (published with the dataset).

AI statement

This codebase and paper were written by Claude Code (Anthropic's Claude, model Claude Opus 5.5) from an exact specification provided by Libor "Burny" Burian, who directed the work. Human cleanup and rewriting are in progress.